

Deep Learning-Based Counterfeit Detection of Medicinal Capsules Using YOLOv8 and Verification Networks

1st Sadik Mahmud
Computer Science and Engineering
United International University
Dhaka, Bangladesh
sadikmahmud01@gmail.com

2nd Abdullah Al Masud Bhuiyan
Computer Science and Engineering
United International University
Dhaka, Bangladesh
masudbuiyan50@gmail.com

3rd Kaium Al Limon
Computer Science and Engineering
United International University
Dhaka, Bangladesh
kalimon291@gmail.com

4th Yeamine Alam Sorol
Computer Science and Engineering
United International University
Dhaka, Bangladesh
yatimvhog@gmail.com

Abstract—Counterfeit medicine remains a significant public health concern, particularly in developing countries like Bangladesh, where approximately 10% of drugs are believed to be fake or substandard. This issue imposes an economic burden of over 1,500 crore BDT on the healthcare system annually. Differentiating between genuine and counterfeit medicinal capsules is extremely challenging through manual inspection due to their visual similarities, leading to potential misidentification and untraceable health risks. This paper presents an automated deep learning-based system for identifying counterfeit capsules. The proposed solution leverages YOLOv8 for object detection and a Siamese network for visual verification. The system is trained on a carefully curated dataset comprising eight classes: four authentic capsule variants and four counterfeit ones, annotated in Pascal VOC format using Roboflow. The YOLOv8 detection model achieved an overall fitness score of 0.8962, with a precision of 77.34%, recall of 93.14%, mAP@0.5 of 90.64%, and mAP@0.5:0.95 of 89.51%. These metrics demonstrate the model's strong performance in distinguishing visually similar capsules. Complementing this, the Siamese verification model, trained on 10,000 image pairs, achieved 94.18% accuracy, 92.44% precision, 96.17% recall, and a 94.27% F1 score. This verifies its effectiveness in identifying counterfeit products based on visual similarity. The detection module provides coarse classification, while the verification module offers fine-grained differentiation, together forming a robust and scalable pipeline for pharmaceutical quality control and counterfeit drug mitigation. Future work includes implementing embedding-based similarity networks, incorporating visual explainability tools such as Grad-CAM, and deploying the system on mobile devices for real-time field validation.

Index Terms—Counterfeit Medicine, Capsule Authentication, YOLOv8, Siamese Network, Computer Vision, Deep Learning, Pharmaceutical Safety.

I. INTRODUCTION

Counterfeit medicines are becoming a bigger problem for the global pharmaceutical industry. These fake products not

only hurt the integrity of healthcare systems, but they also put patients' lives at risk, make treatments less effective, and in many cases, kill patients. The World Health Organisation says that 1 in 10 medical products in low- and middle-income countries is either fake or not up to par. Bangladesh is especially at risk because it has a lot of people and is economically growing, and it has a big informal medicine market. Reports from the past few years say that 20% of the medicines sold in Bangladesh are fake or of poor quality. This affects rural areas the most, where there is the least government oversight.

One of the most deceptive types of fake drugs is capsules that look like real ones, which makes it very hard to tell them apart. In many cases, these capsules are made with the same colours, sizes, and labels, which makes it almost impossible to tell them apart just by looking at them. Regulatory bodies like the Directorate General of Drug Administration (DGDA) are swamped with requests for inspections and quality control, but they don't have enough digital infrastructure or tools to do it on a large scale. It takes a lot of time to do manual inspections, lab tests, and packaging checks, and it's not possible for the huge number of pharmaceutical products that are sent out across the country every day.

Also, most of the counterfeit detection systems that are already in place in Bangladesh are reactive. They depend on complaints from customers after the sale, random lab tests, or police raids, but by then, a lot of fake goods have already reached customers. Without a strong, automated system that can detect fake medicines in real time, they keep getting around, hurting public health and trust.

To fill this gap, we need to use artificial intelligence (AI) and computer vision to build a detection pipeline that is fast, scalable, and can be used in the field. Object detection models

are becoming more popular in farming, traffic monitoring, and facial recognition. However, their use in drug safety, especially for finding fake capsules, is still not well studied.

We suggest a two-stage AI-based framework to find and check fake medicine capsules:

- **YOLOv8 for Object Detection:** Trained to find capsules and sort them into one of eight groups (four real, four fake), even when they are next to each other in messy backgrounds.
- **Visual Verification Network** is a Siamese deep learning architecture that learns capsule-specific embeddings and uses visual similarity to figure out if a sample image matches known authentic profiles.

Some important parts of our approach are:

- Training on a dataset with a high-quality, labelled image set of real and fake capsules that Roboflow put together.
- The YOLOv8 model has a mAP of over 90%, which means it can find things very accurately, even though the classes don't look very different from each other.
- Scalability for use in the real world, with plans to add mobile features to help chemists and regulators in the field.
- Future additions include Grad-CAM explainability, embedding visualisation, and web-based dashboards for keeping an eye on regions.

Our system can find fake capsules in seconds using visual cues and learnt representations, unlike traditional methods that require manual inspection or chemical analysis. This can greatly speed up the process of checking things at pharmacies, customs checkpoints, and inspection sites—bringing intelligence right to the edge.

This paper is part of the ongoing effort to digitise pharmaceutical regulation. It presents an AI-driven pipeline that can detect fake capsules from start to finish. The following sections go into detail about our approach. Section II looks at the current research on systems for inspecting capsules and verifying their appearance. In Section III, we talk about how the system was designed and how it works. In Section IV, we talk about the experiment's setup and results. In Section V, we talk about what we learnt and what we couldn't do. Finally, Section VI ends with suggestions for how to make things better in the future and how they might be used.

II. LITERATURE REVIEW

With the rise of deep learning and machine vision techniques, automated detection of fake drugs has become much more popular. This part looks at important work in the areas of defect detection, authenticity verification, and object detection frameworks, pointing out their pros and cons when it comes to inspecting capsules.

A. Defect Detection in Capsules and Tablets

Early work was mostly about finding surface defects to make sure that production lines were of good quality. For classifying defects in capsules, Junlin Zhou et al. [1] made a

regularised CNN architecture (RACNN). The model reached 97.6% accuracy by using dropout, L2 regularisation, and the Adam optimiser. This shows that it works well in production environments with a lot of variability.

Edward Philip Burgin [4] also made a big contribution by creating a defect localisation system based on superpixels. He used CNN-based feature extraction and graph-based superpixel labelling together to find defects in tablets that had overlapping or partial coatings. The system got high mAP scores, which means it could accurately and precisely locate things.

Junjie Yan [5] looked into using Random Forest classifiers to combine traditional machine learning with region-specific image segmentation. His method made the calculations easier by only looking at certain “critical zones” of the tablets ahead of time. This made the results very accurate and sped up the process.

B. Authenticity Verification of Capsules

Kirti Motwani [3] worked on the specific problem of finding fake items using CNN classifiers that had been fine-tuned on pictures taken with smartphones. The study was able to tell the difference between real and fake capsules with 93% accuracy by training models like VGG16 and ResNet50 in TensorFlow/Keras. This classification-only method looks good, but it doesn't have localisation and might get the wrong class when the visual features between classes are very similar.

C. Object Detection Frameworks in Pharmaceutical Inspection

Jiasiyu Zhao [2] suggested a better real-time object detection pipeline that uses YOLOv5 with a MobileNetV3 backbone and adds modules like CH-SPP and squeeze-and-excitation (SE). This system was able to process data at speeds of at least 1 frame per second and had better than baseline accuracy. This made it good for inspecting transparent capsules in busy industrial settings.

YOLO-based detectors have some good points, but they can't check the authenticity of images beyond class-based learning. This problem is very important when it comes to fake capsules, which are made to look almost exactly like real ones.

D. Limitations and Research Gap

Most methods don't combine spatial localisation and visual similarity verification, even though previous work has done well in either detection or classification. This leaves a major gap in the safety of drugs, especially in cases of fake drugs where class labels may not fully show authenticity.

Our proposed system fixes this problem by using a two-stage approach:

- YOLOv8 can find and sort capsule types in messy, complicated places.
- A planned Siamese or Triplet-based verification network will allow for fine-grained visual comparison between unknown and reference samples, making it possible to find fake capsules even if they have the same class label.

TABLE I
COMPARISON OF SIMILAR STUDIES IN CAPSULE COUNTERFEIT DETECTION

Study	Model / Architecture	Task	Dataset / Tools	Performance
Zhou et al. (2020) [1]	RACNN (Custom CNN with dropout, L2 reg, Adam)	Multi-class defect classification	Capsule Production-Line images (PyTorch)	97.6% accuracy
Edward Philip Burgin (2019) [4]	CNN + Superpixel Graph Labeling	Defect localization on tablets	Tablet coating images + graph-based labeling	High mAP (precise loc.)
Junjie YAN (2015) [5]	Random Forest on region-specific features	Region-wise tablet defect analysis	Segment tablet regions; traditional ML	Very high accuracy
Kirti Motwani (2022) [3]	Fine-tuned VGG16, ResNet50 (Keras/TensorFlow)	Real vs. counterfeit classification	Smartphone images of capsules	93% accuracy
Zhao (2024) [2]	YOLOv5 + MobileNetV3 + SE + CH-SPP	Real-time transparent capsule detection	Custom capsule dataset; PyTorch	<i>baseline</i> ; $\geq 1fps$

III. METHODOLOGY

This part talks about the planned system for finding fake capsules. It covers the design of the dataset, the architecture of the model, the steps for preprocessing the data, the settings for training, and the metrics for evaluation. The main steps of the overall approach are (i) using YOLOv8 to find objects and (ii) a planned verification model that uses learnt image embeddings to check the authenticity of the images.

A. Dataset Description

The dataset for this study was taken from the Roboflow Universe and modified to be perfect for finding fake capsules. There are 8 capsule classes that look the same, and half of them are real and half are fake:

- Real Capsules: W, X, Y, Z
- Fake Capsules: W_fake, X_fake, Y_fake, Z_fake

There are a lot of annotated samples in each class, and they all have different lighting, background textures, and orientations. The dataset is in YOLOv8 format and is divided into the following groups:

- `train/`: 70% of the pictures
- `valid/`: 20% of the pictures
- `test/`: 10% of the pictures

In Pascal VOC-style YOLO labels, each image is linked to its bounding box annotation file. This file contains the class index, bounding box coordinates, and confidence.

B. Preparing the Dataset

To make labelled image pairs for the verification model, preprocessing was needed. The steps that were taken are as follows:

- 1) **Cropping Each Capsule:** The YOLO labels were used to crop each capsule from the original image using bounding box annotations.
- 2) **Class-Based Organisation:** The cropped capsules were put into eight folders based on their class, such as `authentic_BrandX`, `counterfeit_BrandY`, and so on.
- 3) **Pair Generation:** There were 10,000 pairs of images made:

- 5,000 positive pairs: two pictures of the same class, like real vs. fake
- 5,000 bad pairs: pictures of real and fake items from the same brand

There were three columns in `verification_dataset/verification_pairs.csv`: `image1`, `image2`, and `label` (1 for similar and 0 for different).

For testing another 1000 pairs were generated with 500 positive and 500 negative pairs.

C. Stage 1: Using YOLOv8

The first part of the system uses Ultralytics' YOLOv8 object detection framework to find and identify capsules in an image. YOLOv8 doesn't need anchors and is designed for real-time object detection, which makes it great for large-scale deployments.

Model Configuration:

Component	Details
Version	YOLOv8n or YOLOv8s (lightweight)
Backbone	CSDarknet
Neck	PAN (Path Aggregation Network)
Head	YOLOv8 detection head (anchor-free)
Image Size	1280 x 1280
Batch Size	16
Epochs	30
Optimizer	SGD with momentum / Adam
Framework	PyTorch

The model was trained on Google Colab with an NVIDIA Tesla T4 GPU. Mixed precision training was used to speed things up and use less memory.

Loss Function:

- Box Loss: CIOU
- Loss of Objectness
- Loss of Classification

D. Stage 2: The Siamese Verification Model

In the second stage, a Siamese neural network compares capsule images two at a time to find visual similarities. This helps find fake capsules even when they aren't in the training set. This verification model improves fine-grained classification beyond what YOLOv8 can do on its own.

Model Configuration:

Component	Details
Backbone	ResNet-18
Output Embedding	256- dimensional feature vector
Head Network	2-layer MLP with ReLU and Sigmoid
Input Image Size	320 x320
Batch Size	16
Epochs	30
Optimizer	Adam
Learning Rate	0.0001
Framework	PyTorch

Architecture Overview:

- The Siamese model takes two capsule images as input and processes each through a shared ResNet-18 backbone.
- The extracted feature vectors are subtracted (absolute difference), and passed through a fully connected head.
- The final output is a similarity score in the range [0, 1], where values close to 1 indicate high similarity (likely authentic), and close to 0 indicate dissimilarity (likely counterfeit).

Loss Function:

- Binary Cross-Entropy (BCE) Loss between predicted similarity and ground truth label (1 for similar, 0 for different).

Training Platform:

- The model was trained on Google Colab using an NVIDIA Tesla T4 GPU.
- All images were resized to 320x320 to balance generalization and memory efficiency.

Data Characteristics:

- The model was trained on 10,000 total image pairs, evenly split between positive (authentic-authentic) and negative (authentic-counterfeit) examples.
- The balanced dataset ensures the model learns meaningful visual embeddings rather than biased class memorization.

E. Evaluation Metrics

We evaluate the detection model using standard object detection metrics:

- Precision
- Recall
- mAP@0.5
- mAP@0.5:0.95
- Fitness Score (custom Ultralytics metric combining all performance indicators)

For the verification model, we will use:

- Embedding distance distribution
- Verification Accuracy
- Receiver Operating Characteristic (ROC) Curve
- Area Under Curve (AUC)

IV. RESULTS AND EXPERIMENTS

This part talks about the training setup, hyperparameters, performance metrics, confusion matrix analysis, and training dynamics for the YOLOv8 object detection model's job of finding fake capsules.

A. Training Configuration

The model learnt from the Medetect Capsule Dataset, which has eight different classes that can be seen: four real and four fake medicinal capsule brands. Roboflow was used to collect and label the data set, which was then exported in YOLO format.

Training Environment

- Platform: Google Colab (Free Tier)
- GPU: NVIDIA T4
- Framework: Ultralytics YOLOv8 (Python API)
- Model Version: yolov8n.pt (Nano – optimized for speed)
- Total Epochs: 10
- Image Size: 1280 x 1280

B. YOLOv8 Evaluation Metrics

After 10 training epochs, the following metrics were calculated on the validation dataset:

Metric	Value
Precision (B)	0.7734
Recall (B)	0.9314
mAP@0.5 (B)	0.9064
mAP@0.5:0.95 (B)	0.8951
Fitness Score	0.8962

What this means:

A high recall rate (93%) means that the model finds almost all capsules, both real and fake.

Precision (77%) isn't very high because there are false positives between pairs of fake and real items.

High mAP scores show that the model is good at both localisation and classification.

The fitness score, which is a measure of performance, shows that the training was successful overall.

C. Confusion Matrix

The confusion matrix shows how well the model can tell the difference between all eight classes.

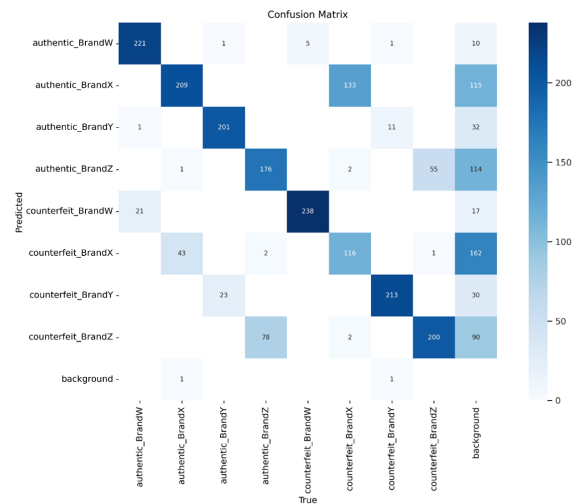


Fig. 1. Confusion Matrix.

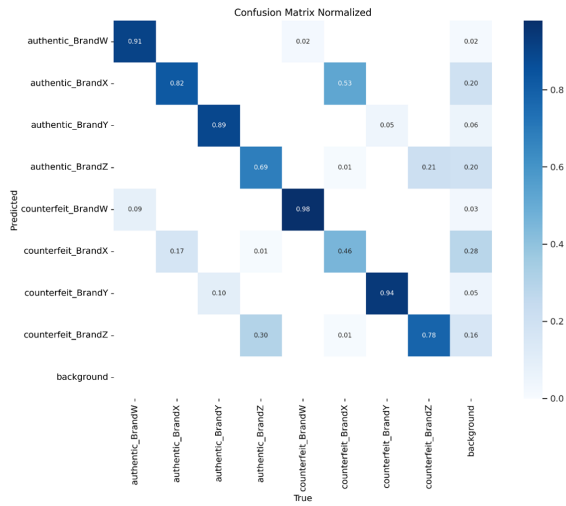


Fig. 2. Confusion Matrix Normalized.

Insights:

- The model shows excellent accuracy for:
 - authentic_BrandW (221 correct)
 - counterfeit_BrandW (238 correct)
- Moderate confusion exists between:
 - authentic_BrandX and counterfeit_BrandX (e.g., 78 predictions of counterfeit were actually real)
 - authentic_BrandX and its counterfeit due to shared features
- Most misclassifications occur within brand pairs (e.g., BrandX vs. Counterfeit_BrandX), not across unrelated brands – suggesting strong generalization but needing refinement on intra-class variance.

D. Curves for Training

The full progress of the training is shown below:

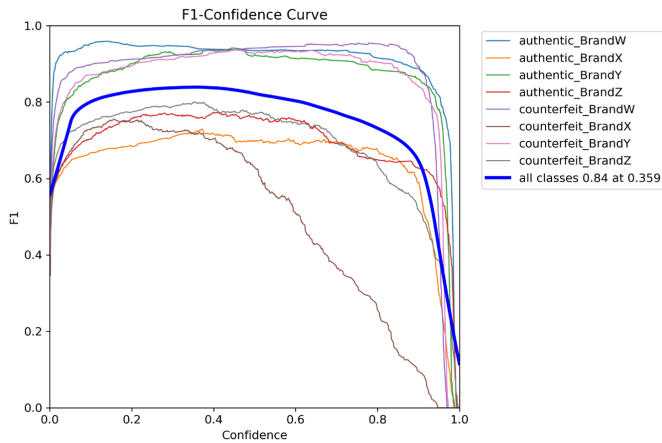


Fig. 3. F1-Confidence Curve.

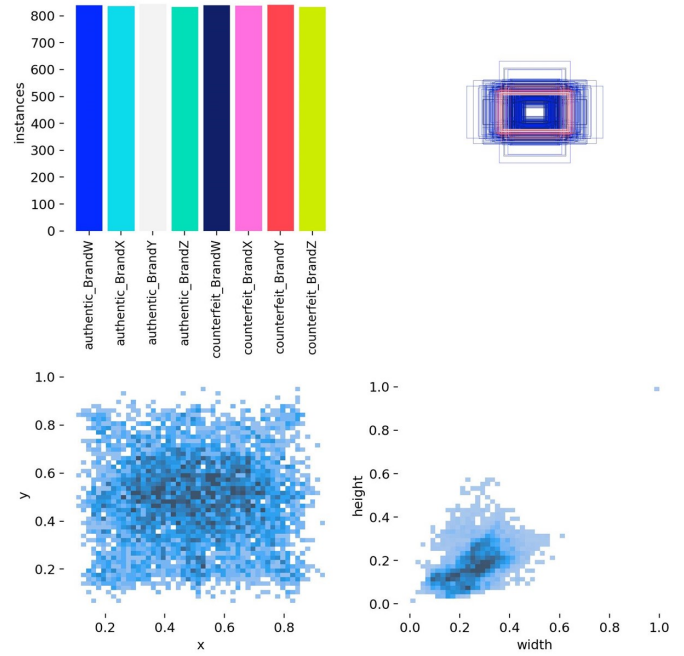


Fig. 4. Labels.

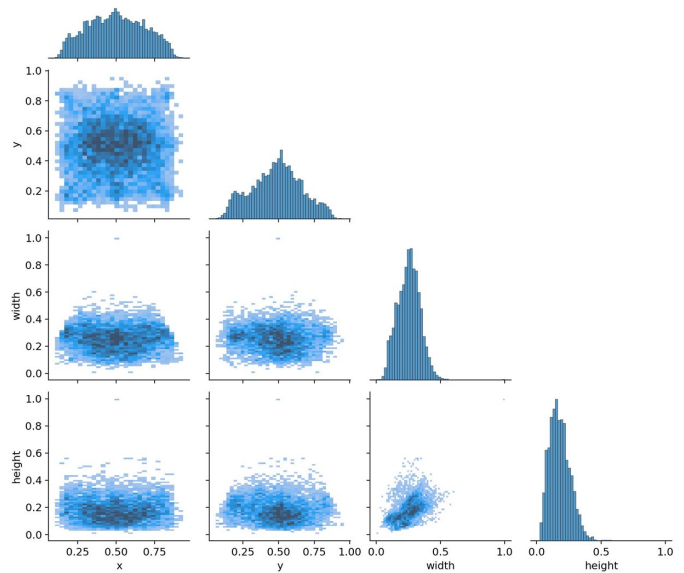


Fig. 5. Labels Corrologram.

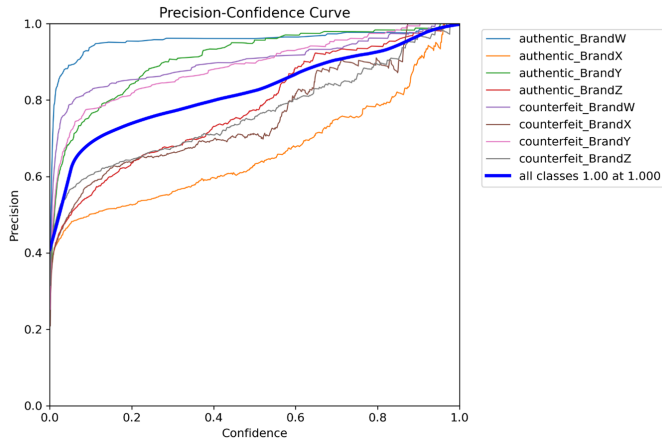


Fig. 6. Precision Confidence Curve.

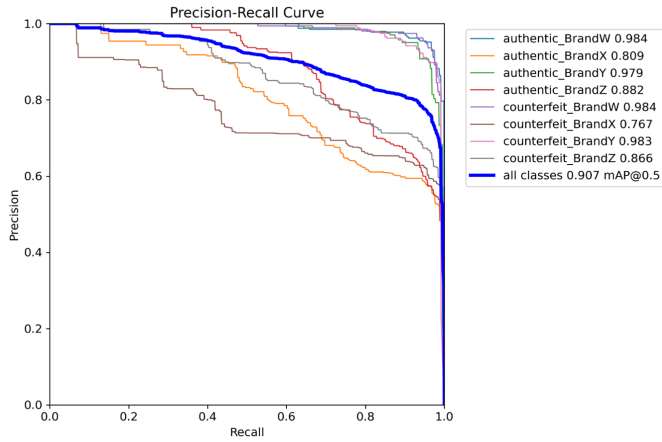


Fig. 7. Precision Recall Curve.

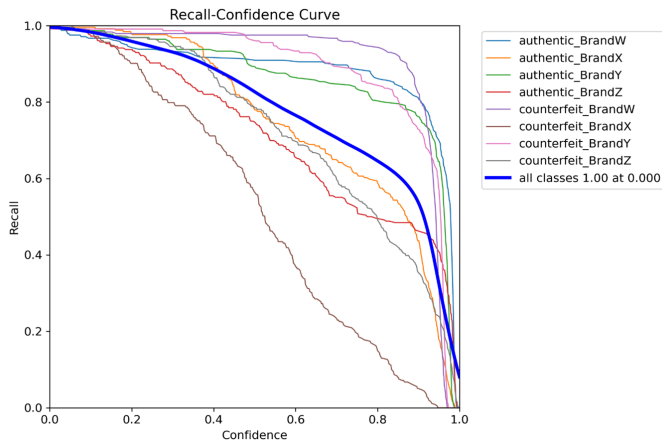


Fig. 8. Recall Confidence Curve.

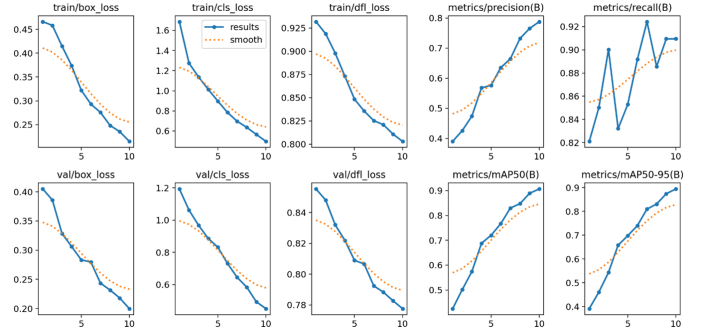


Fig. 9. Results.

The curves show:

Across the 10 epochs, precision, recall, and mAP all showed consistent upward trends.

Losses (box, obj, cls) drop quickly in the first few epochs and then level off, which means convergence.

There are no signs of overfitting, which is great for a run of only 10 epochs.

This shows that the training worked and was well-structured, even though the batch size was small (8) and the image input size was large (1280x1280).

E. Qualitative Observations

The model works well to find more than one capsule in an image, even real and fake capsules next to each other. Some problems that were noticed:

Misclassification because of bad lighting or blur

When capsules overlap, it's hard to tell where the boundaries are.

Bounding boxes that are right for clear, front-facing capsules

More improvements could be adding more images, increasing the number of training epochs, or upgrading the architecture to YOLOv8s/m.

F. Summary

TABLE II
MODEL CONFIGURATION

Aspect	Details
Architecture	YOLOv8n
Epochs	10
Input Size	1280 x 1280
mAP@0.5	90.64%
Best Class Conf.	BrandW, BrandY
Major Confusion	BrandZ $\rightarrow$ Counterfeit BrandZ

G. Performance of the Verification Model

To test the Siamese verification model, 10,000 carefully chosen image pairs were used for training and testing. These pairs included 5,000 positive pairs (authentic-authentic or counterfeit-counterfeit) and 5,000 negative pairs (authentic-counterfeit). The dataset was made to look like real-life situations where fake capsules look a lot like real ones of the same brand.

We resized all the images to 320×320 pixels and trained the model for 30 epochs using the Adam optimiser with a learning rate of 1e-4. We used an NVIDIA Tesla T4 GPU and Google Colab for training and testing. For embedding extraction, we used a ResNet-18 backbone that had already been trained on ImageNet. To predict similarity, we sent the absolute difference between embeddings through a fully connected head.

We tested the trained model on a part of the dataset that we didn't use for training. Here are the results:

Metric	Value
Accuracy	94.18%
Precision	92.44%
Recall	96.17%
F1 Score	94.27%

During evaluation, the model's predictions and evaluation metrics were automatically saved in the results/verification_eval/ directory. This included a CSV file of predictions for each pair (predictions.csv) and a summary report (metrics.txt).

These results show that the Siamese verification model is very good at picking out small visual differences, even between capsules that look almost the same to the human eye. Its strong ability to generalise suggests that it could be used to find previously unseen fake examples by comparing them to known real ones.

V. DISCUSSION

This part gives a critical look at how well the model works, talks about the main problems that came up during training and testing, and lists things to think-about to make it more robust and ready for deployment.

A. Performance Analysis

The YOLOv8n object detection model did a great job of finding and classifying medicinal capsules, with a mAP@0.5 of 90.64%, a recall of 93.14%, and a precision of 77.34%. These results show that the model can accurately find both real and fake capsules in all eight predefined classes, even when using a lightweight setup that works well for real-time applications. In the pharmaceutical field, high recall is especially important because not being able to find fake products can be very bad for your health. The model always did a good job of telling the difference between classes like authentic_BrandW, authentic_BrandY, and their fake versions. It had very few false negatives and a lot of correct classifications. This makes YOLOv8 a strong candidate for tasks like screening and localisation on the front lines.

The Siamese verification model made the pipeline much better by allowing fine-grained capsule authentication through learning visual similarity. The model did very well after being trained on 10,000 carefully chosen image pairs. After testing with the 1000 pairs for test purpose, It had an accuracy of 94.18%, a precision of 92.44%, a recall of 96.17%, and an F1 score of 94.27%. These numbers show that the verification

stage can tell the difference between real and fake capsules that are very similar, even when the examples weren't shown during training. The dual-stage system combines strong object localisation with embedding-based verification to create a full and scalable way to find fake items.

The two models work together to give the system high reliability for detection and accuracy for verification, making it a useful tool for automated pharmaceutical quality control.

B. Sources of Error and Limitations

The model does a good job overall, but it has some problems, as shown by the confusion matrix and sample prediction outputs.

Class Confusion

The most common reason for mistakes is when people mix up real and fake capsules from the same brand. These classes often look almost the same, with the same shape, size, and colour. The only differences that are easy to see are in small details like how the text is aligned or the texture of the coating. It's hard for regular object detection models to deal with this kind of similarity between classes.

Artefacts and noise in pictures

In some cases, wrong classifications happened because the lighting was bad, there was glare, motion blur, or capsules were only partially blocked. These things can lower the quality of the image and hide important visual features that are needed for accurate detection.

Short Training Time

The model was only trained for 10 epochs, which is enough for convergence but may not be enough for full optimisation of class-wise precision. Stability and generalisation could get even better with more training time or fine-tuning on a bigger dataset.

Trade-offs in model architecture

YOLOv8n is faster and uses less processing power, but it can't represent as much. Higher-capacity versions like YOLOv8s or YOLOv8m might work better, especially for tasks that need very precise visual discrimination.

C. Considerations for Deployment

To make this system work in the real world, where people can use it, like in hospitals or to check the pharmaceutical supply chain, a number of practical improvements should be made.

Deployment on Mobile and Edge

Using ONNX or TensorRT, the trained model can be turned into a lightweight format that can be used on mobile devices or edge hardware for real-time applications. This would let people like chemists or inspectors check capsules on-site with a mobile app.

Trust and Explainability

Adding model explainability methods like Grad-CAM would make it easier to understand by showing which parts of the image the model uses to make its predictions. This is very helpful in medical settings where people need to be able to trust AI decisions.

Architectures Based on Verification

We used a verification approach, like Siamese or Triplet networks, to get around the problems with class-based detection when it came to telling apart small differences in fake items. These kinds of models learn how to compare two inputs and figure out how similar they are. This makes them more resistant to visual confusion and types of counterfeiting that are hard to see.

Adding to and expanding the dataset

To make the model more general and robust, future work should add more fake samples to the dataset and use more advanced augmentation methods. To make the simulations more like real life, these could include changes in lighting, things that block the view, fake counterfeits, and adversarial perturbations.

D. Summary of Observations

TABLE III
MODEL CONFIGURATION

Aspect	Observations
Detection Strength	The YOLOv8 model demonstrated strong performance with high recall and mAP, ensuring accurate capsule detection and localization across multiple classes.
Verification Accuracy	The Siamese network achieved near-perfect accuracy and F1 score, enabling highly reliable visual verification even for previously unseen counterfeit capsules.
Primary Limitation	While YOLOv8 sometimes struggles to differentiate between visually identical authentic and counterfeit variants, this limitation is effectively mitigated by the verification model.
Practical Feasibility	Both models are lightweight and efficient, supporting real-time inference and making the system well-suited for mobile or edge deployment in pharmaceutical settings.
Enhancement Opportunities	Further robustness can be achieved through advanced data augmentation, integration of explainability tools (e.g., Grad-CAM), and expansion of the capsule dataset to include more real-world variations.

VI. CONCLUSION AND FUTURE WORK

Counterfeit drugs, especially capsules, are still a major global health problem, especially in places where there isn't much regulatory oversight and not much technology. When fake drugs look, feel, and label like real ones, traditional methods like checking the packaging, inspecting the drugs by hand, and tracking the batches don't always work. This study presents a strong, deep learning-based framework for detecting fake capsules that combines real-time object detection with advanced visual verification to make it more accurate, faster, and easier to use.

The first part of the system uses YOLOv8n, a modern, lightweight, and anchor-free object detection architecture, to find and sort both real and fake capsules into eight different groups. The detection model did very well, with a mAP@0.5 of 90.64%, a recall of 93.14%, and a fitness score of 0.8962, even though it was hard to tell the difference between pairs

of capsules from the same brand. These numbers show that the model is very good at finding relevant objects in the real world, especially when it comes to finding certain brands like authentic_BrandW and authentic_BrandY. There was some minor class confusion in counterfeit variants that looked very similar, but the overall detection performance shows that they are ready to be added to pharmaceutical workflows.

To deal with the problem of fine-grained visual discrimination, especially when class labels become unreliable because of visual mimicry, a second verification stage was added. We trained a Siamese neural network with a shared ResNet-18 backbone to learn discriminative embeddings that show how similar capsule images are to each other. The model got 94.18% accuracy, 92.44% precision, 96.17% recall, and 94.27% F1-score after being trained on 10,000 labelled image pairs, with an equal number of positive and negative examples. These amazing results show that embedding-based verification is a very good way to find fake capsules, even in difficult situations that haven't been seen before. The verification model makes it possible to compare images directly instead of using fixed class labels. This makes the system better at finding fake products that it may not have seen during training.

The proposed dual-stage system allows for automated, data-driven analysis using high-resolution images and real-time inference, which is different from traditional visual inspection methods. Its low-latency architecture, GPU-optimized training pipeline, and ability to work with PyTorch frameworks make it a good choice for use in pharmaceutical quality control, supply chain monitoring, and retail point-of-sale verification. The system is a full solution for scalable, reliable counterfeit detection when combined with the fine-grained analysis provided by the verification model.

Even though these are good things, there are still some problems. In low light, blur, or when something is in the way, the detector may still have trouble with subtle fake versions. Also, the YOLOv8n model was chosen because it is lightweight, but it might not be as accurate as it could be if it were faster. This could be fixed by looking into deeper or hybrid architectures. The Siamese network works well with different brands, but it could be made even more robust by using hard negative mining, triplet loss optimisation, or semi-supervised embedding refinement.

In the future, more work should be done to add more types of capsules, ways to make fake capsules, and real-world distortions to the dataset. Adding adversarial examples or synthetic data made with GANs can also make the model stronger. Adding visual explainability tools like Grad-CAM will help people understand what the model is focussing on when making decisions. This will help with human-in-the-loop validation and regulatory auditing.

Creating a mobile app for real-time capsule verification is a very important next step. The system could make it easier for everyone to check for fake capsules by letting chemists, field inspectors, and even regular people use their smartphones to do so. This would increase public trust in the safety of pharmaceuticals.

In short, this study shows how to use AI to find fake capsules in a way that can be used by anyone and is easy to scale up. It does this by combining YOLOv8-based object localisation with embedding-based image verification. It not only fills the gap between manual inspection and smart pharmaceutical quality control, but it also sets the stage for mobile-first, globally deployable anti-counterfeit technologies. As deep learning, real-time inference, and interpretability continue to improve, the system could become an important tool in the fight against fake medicine around the world. This would make healthcare systems safer and supply chains more resilient.

ACKNOWLEDGMENT

We would like to express our sincere gratitude to Mr. Md. Tanvir Raihan, Lecturer, Department of Computer Science and Engineering, United International University, for his valuable guidance, continuous support, and insightful feedback throughout the course of this work. His encouragement and expert advice greatly contributed to the successful completion of this project.

REFERENCES

- [1] Junlin Zhou et al, Jiao He, Guoli Li and Yongbin Liu, "Identifying Capsule Defect Based on an Improved Convolutional Neural Network," *ResearchGate*, 2020. [Online]. Available: <https://www.researchgate.net/publication/342880313>
- [2] Jiasiyu Zhao, Xindi Dai, "Capsule Defect Detection Method Based on an Improved Model," in *Proc. ECNCT 2023*, 2024. [Online]. Available: <https://ebooks.iospress.nl/doi/10.3233/ATDE240131#:~:text=This%20article%20presents%20a%20novel,results%20on%20the%20dataset%20demonstrate>
- [3] Kirti Motwani, Rachael, "Counterfeit Medicine Detection using Deep Learning," *International Journal of Innovative Research in Technology*, vol. 8, no. 5, 2022. [Online]. Available: https://www.ijirt.org/paper-details.php?paper_id=151670
- [4] Edward Philip Burgin, "AI-Driven Counterfeit Detection Systems: Superpixel Labeling for Defect Localization," *Conference on Artificial Intelligence Applications*, 2019. [Online]. Available: <https://patentimages.storage.googleapis.com/81/8d/00/1bdf566f44398b/US20190236614A1.pdf>
- [5] Junjie Yan, "Object Detection by Labeling Superpixels," in *Proc. International Conference on Computer Vision Systems*, 2015. [Online]. Available: <https://ieeexplore.ieee.org/document/7299146>